Foxglove Integration Options for Dataset Hub

Dataset Hub, 23 Oct 2026. Covers Foxglove and Lichtblick, the open-source fork of Foxglove that takes the same URLs.

There are three ways to integrate Foxglove, from least to most effort. The README already lists "dataset preview via Lichtblick" as an open point. Option 1 covers both viewers.

1. "Open in Foxglove" button per catalog row (recommended start)

Foxglove can stream a remote MCAP file straight from a URL:

```
https://app.foxglove.dev/~/view?ds=remote-file&ds.url=<url-to-file.mcap>
foxglove://open?ds=remote-file&ds.url=<url>     # desktop app deep link
```

The viewer reads only the byte ranges it needs, so large bags open fast without a download. The hub has to do three things for this to work:

- **Serve bag files with HTTP Range and CORS.** Add `GET /files/<dataset_id>/<name>` to `hub/server.py`. It must handle the `Range:` header (Python's stdlib file handler doesn't), return `206 Partial Content` and `Accept-Ranges: bytes`, and send `Access-Control-Allow-Origin`. About 40 lines of stdlib code.
- **Make sure there's an MCAP to point at.** Remote-file mode supports `.mcap` and ROS1 `.bag`, not rosbag2 `.db3`. During ingest, if a bag isn't MCAP, generate a `preview.mcap` next to it with `mcap convert` or `ros2 bag convert`. The originals stay untouched, which matches the `copy` philosophy. A background job queue (already on the next-steps list) is a natural place for this.
- **Add the button** in `static/index.html`. Building the URL from the catalog row is trivial.

Watch out for mixed content. app.foxglove.dev is HTTPS. Browsers let it fetch from `http://127.0.0.1`, but they block plain-HTTP LAN addresses like `http://192.168.x.x:8080`. Once colleagues open bags on the workstation over the network, you need one of these:

- **Self-host Lichtblick from the hub itself** (serve its static build under `/viewer/`). Viewer and files share an origin, so there's no CORS, no mixed-content problem, no cloud and no license question. This fits an on-prem, no-auth tool best.
- Put TLS in front of the hub (Caddy or nginx).
- Use the Foxglove desktop app with `foxglove://` links.

2. Make the viewer aware of your metadata

- **Default layouts per robot or sensor set.** Store layout JSON files keyed by `robot` or `sensors` (e.g. in `layouts/`) and pass a layout with the link, so the vineyard robot opens with its camera, lidar and map panels already arranged. In Foxglove you'd use org-shared layouts; check whether your plan supports picking one via a URL parameter.
- **Embed `dataset.json` in the preview MCAP.** MCAP has native `Metadata` records. When writing `preview.mcap`, add customer, project, site, sensors and so on, so any MCAP tool sees the context even when the file is passed around without the catalog.
- **Custom panel extension** (works in Foxglove and Lichtblick). A small panel that calls `/api/datasets/<id>` and shows or edits the metadata, or sets the `quality` flag from the open points, while someone watches the replay.

3. Foxglove Data Platform (heavier, overlaps with what exists)

Foxglove's paid platform has its own recording storage, devices, time-range queries, and an **Events** API for tagging time segments. The hub could sync into it: robot becomes a device, each dataset becomes an import, and quality or replay-job results become events. Foxglove also offers self-hosted storage sites. Before going this way, weigh:

- It duplicates the catalog, which `schema.py` is the source of truth for.
- Customer data (`customer_a/...`) may not be allowed in a third-party cloud.
- It's a recurring license cost.

It only makes sense if you want Foxglove's cross-recording time-range search and events and are happy to treat the hub as the ingest and metadata front end.

Recommendation

Do option 1 with self-hosted Lichtblick: Range-capable `/files` endpoint, `preview.mcap` for non-MCAP bags, and an "Open" button. Add per-robot layouts after that. It closes the README's open point, needs no cloud or license, and the same links work in Foxglove if it's adopted later.

Note: the exact URL parameters have not been checked against current Foxglove and Lichtblick docs; confirm them when building.